EE-222 Data Structures and Algorithms

BS(CE)-2k21

LAB REPORT # 1

Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	**5**	**5**	**5**	**15**
Obtained				

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Lab Instructor.
Kaynat Rana

Experiment # 1

Operations on Linear Arrays

Objective:
The objectives of this session are to understand the various operations on arrays in C++.

Equipment Used:
Dev C++/Visual studio

Procedure:
Operations on linear arrays:
A data structure is a particular way of organizing data in a computer so that it can be used effectively. The idea is to reduce the space and time complexities of different tasks.

Traversing linear arrays:

Traversal in a linear array is the process of visiting each element once. Traversing is done by starting with the first element of the array and reaching to the last. Traversing operation can be used in counting the array elements, printing the values stored in an array, updating the existing values or summing up all the element values.

Algorithm of traversal linear arrays:

1. [Initialize counter] Set K=LB.
2.Repeat Step 3 and 4 while K<=UB.
3.[Visit element] Apply PROCESS to LA[K].
4.[Increase Counter] Set K=K+1.
End of step 2 loop
5.Exit

Inserting Linear Arrays:

In this case we have to move all the elements one position backwards to make a hole at the beginning of array. Through the insertion process is not difficult but freeing the first location for new element involves movements of all the existing elements.

Algorithm of inserting linear arrays:

1.[Initialize Counter] Set J=N.
2.Repeat step 3 and 4 while J>=loc.
3.[Move Jth element downward] Set LA[J+1] =LA[J].
4.[Decrease Counter] Set J=J-1.
5.[Insert element] Set LA [loc]=ITEM.
6.[Reset c] Set c=c+1.
7.Exit

Deletion Linear Arrays:

Deletion refers to a moving an existing element from the array and re-organizing all the element of an array.

Algorithm of Deletion Linear Arrays:

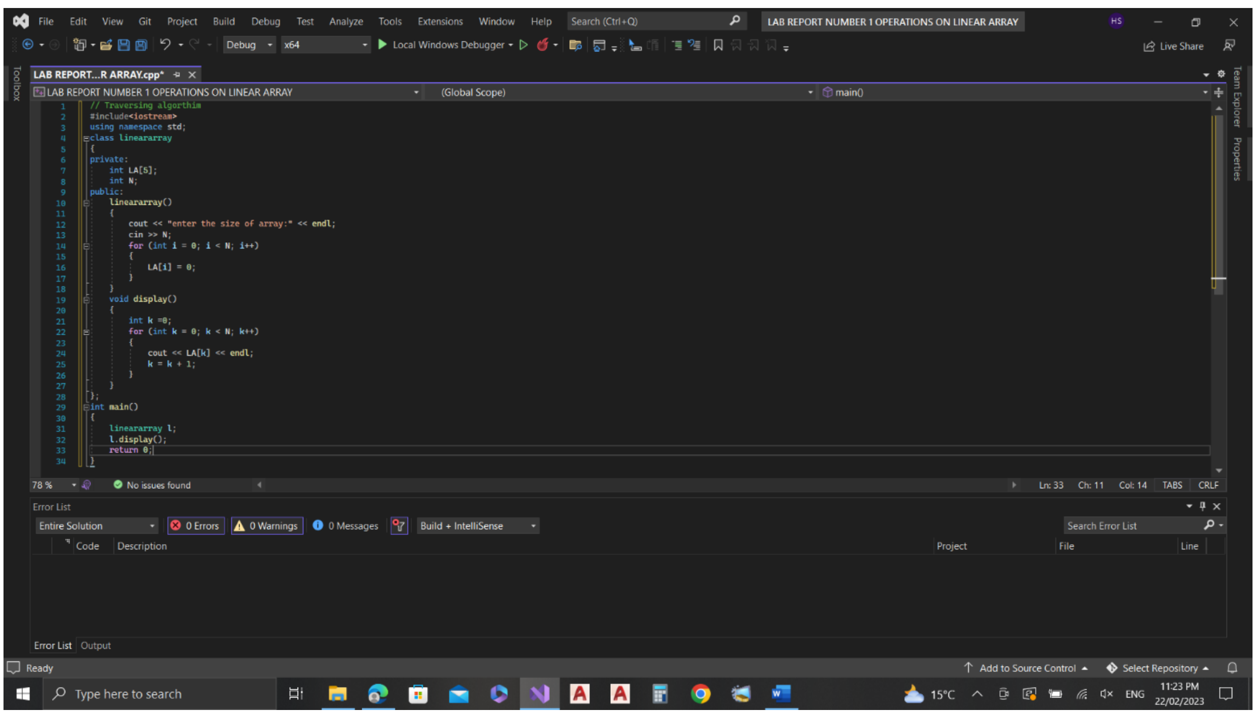
1.[Initialize Counter] Set J=K.
2.Repeat Step 3 and 4 while J<N-1.
3.[Move Jth element upward] Set LA[J] = LA[J+1].
4.[Increase Counter] Set J=J+1.
5.End of Step 2 Loop.
6.[Reset c] Set c=c-1.
7.Exit.

Tasks

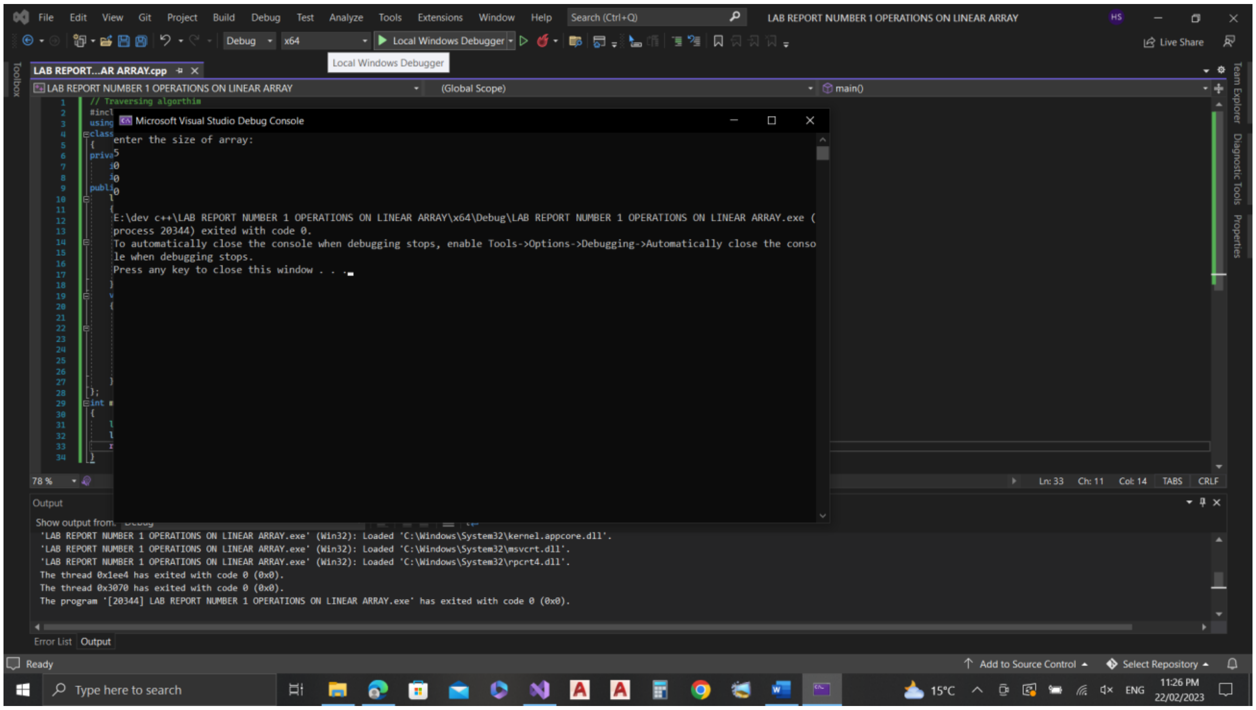
Task 1:

Write a C++ program to implement all the above-described algorithms and display the output for each.

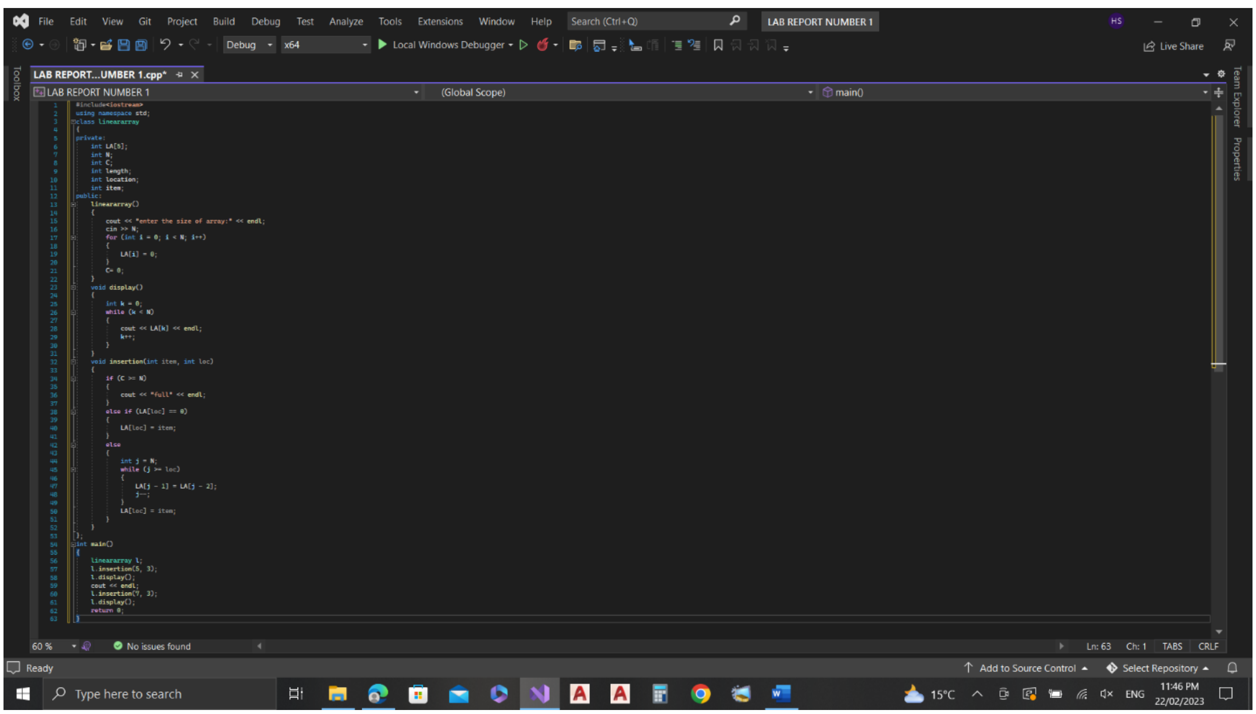
CODE:



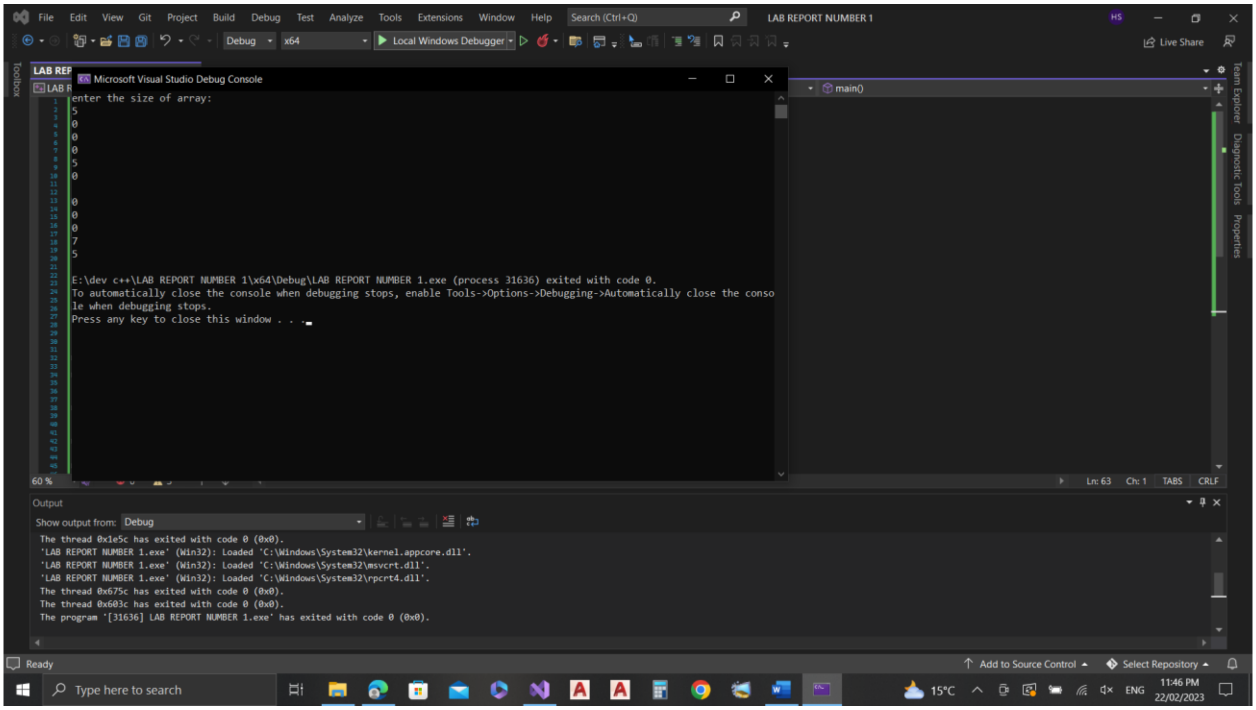
OUTPUT:


Microsoft Visual Studio Debug Console
enter the size of array:
5
0
0
0
E:\dev c++\LAB REPORT NUMBER 1 OPERATIONS ON LINEAR ARRAY\x64\Debug\LAB REPORT NUMBER 1 OPERATIONS ON LINEAR ARRAY.exe (process 20344) exited with code 0.
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
'LAB REPORT NUMBER 1 OPERATIONS ON LINEAR ARRAY.exe' (Win32): Loaded 'C:\Windows\System32\kernel.appcore.dll'.
'LAB REPORT NUMBER 1 OPERATIONS ON LINEAR ARRAY.exe' (Win32): Loaded 'C:\Windows\System32\msvcrt.dll'.
'LAB REPORT NUMBER 1 OPERATIONS ON LINEAR ARRAY.exe' (Win32): Loaded 'C:\Windows\System32\rpcrt4.dll'.
The thread 0x1ee4 has exited with code 0 (0x0).
The thread 0x3070 has exited with code 0 (0x0).
The program '[20344] LAB REPORT NUMBER 1 OPERATIONS ON LINEAR ARRAY.exe' has exited with code 0 (0x0).

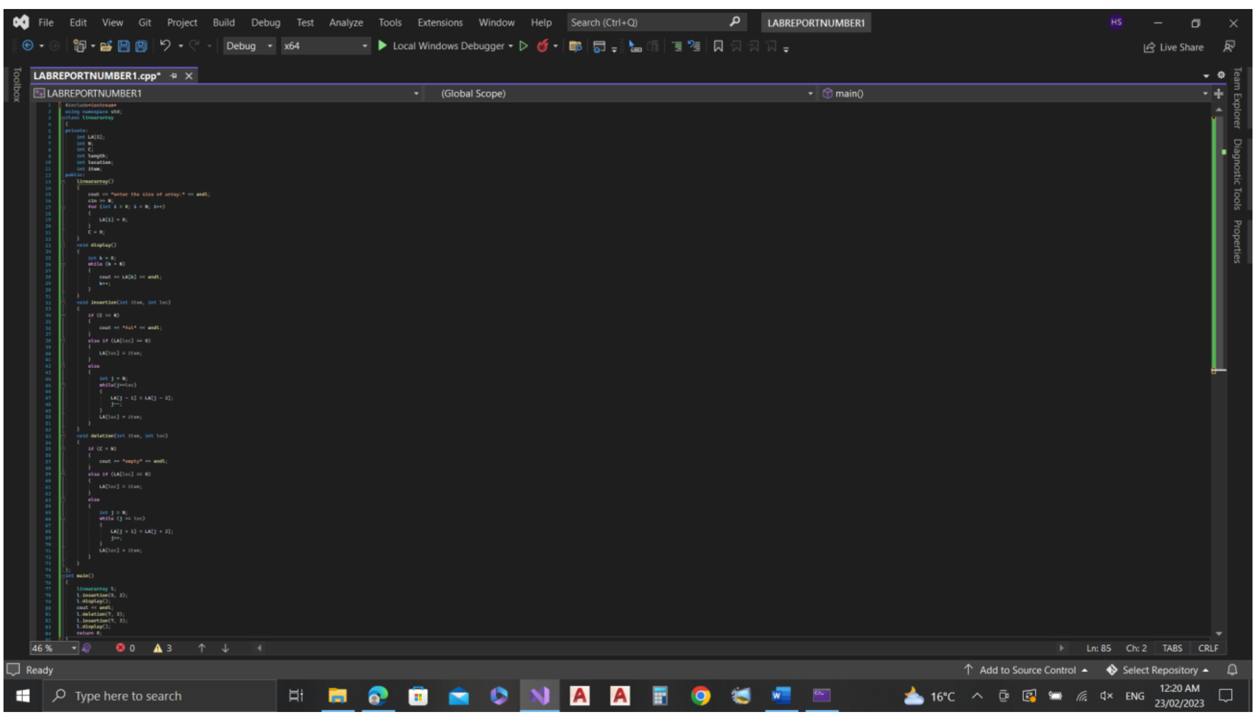
CODE:



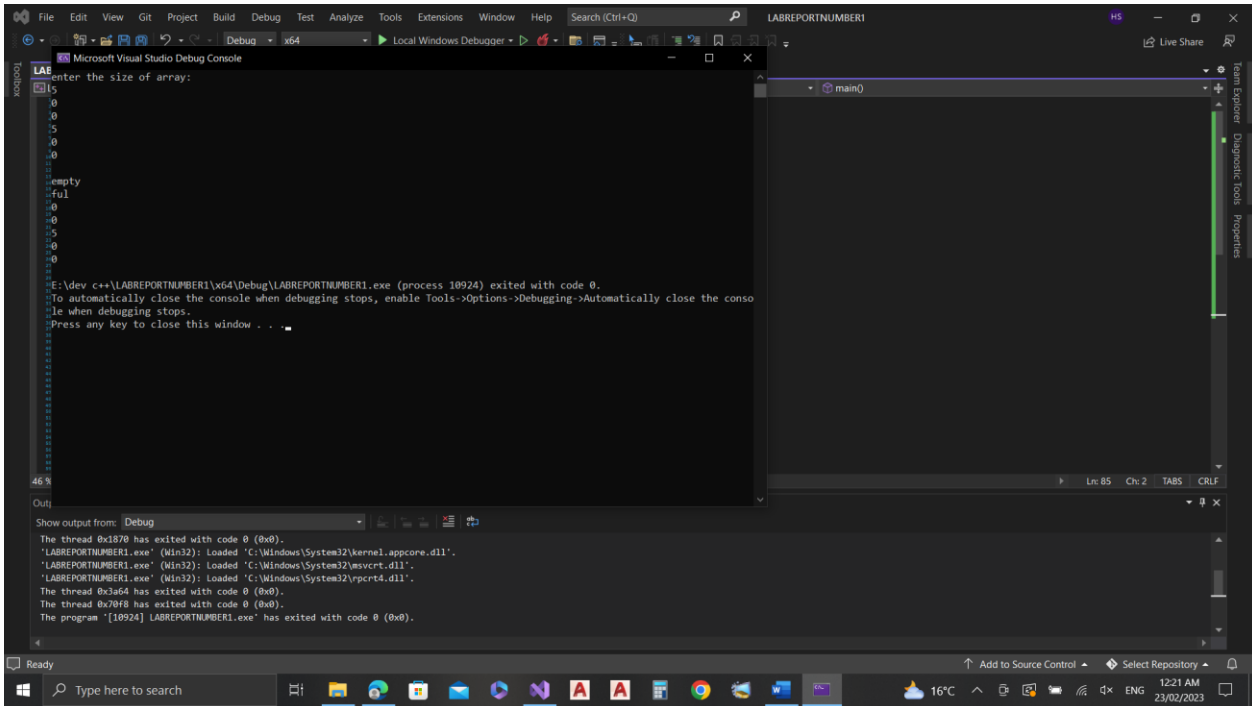
Output:


Microsoft Visual Studio Debug Console
enter the size of array:
5
0
0
0
5
0
0
0
0
7
5
E:\dev c++\LAB REPORT NUMBER 1\x64\Debug\LAB REPORT NUMBER 1.exe (process 31636) exited with code 0.
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
Output
Show output from: Debug
The thread 0x1e5c has exited with code 0 (0x0).
'LAB REPORT NUMBER 1.exe' (Win32): Loaded 'C:\Windows\System32\kernel.appcore.dll'.
'LAB REPORT NUMBER 1.exe' (Win32): Loaded 'C:\Windows\System32\msvcrt.dll'.
'LAB REPORT NUMBER 1.exe' (Win32): Loaded 'C:\Windows\System32\rpcrt4.dll'.
The thread 0x675c has exited with code 0 (0x0).
The thread 0x603c has exited with code 0 (0x0).
The program '[31636] LAB REPORT NUMBER 1.exe' has exited with code 0 (0x0).

CODE:


LABREPORTNUMBER1.cpp*
LABREPORTNUMBER1
(Global Scope)
main()

OUTPUT:



Conclusion:

We use the equipment the visual studio C++ for the operations on linear array. A thorough understanding of these operations and their properties is essential for designing effective algorithms to solve problems.